

HealthRiskAI: Multi-Disease Risk Prediction System Using Machine Learning

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Abstract

Diseases that affect one's health include kidney diseases, heart diseases, and diabetes. These diseases are among the leading causes of death in the world. This paper introduces a system called HealthRiskAI, which employs machine learning methods for predicting risks from health data on multiple diseases. The system makes use of three different datasets (Kidney, Heart, and Diabetes). It uses the process of data preprocessing, feature selection, and classification techniques for creating predictions. The deployment of the model involves an application that can take input parameters and provide instantaneous predictions.

Keywords—Health Prediction, Machine Learning, Disease Risk, Kidney Disease, Heart Disease, Diabetes, AI in Healthcare

1. Introduction

Some of the diseases which contribute to mortality rates include diabetes, cardiovascular disease, and chronic kidney disease. These diseases are characterized by slow onset, with some individuals even failing to realize their severity until their late stages. For this reason, predicting their emergence can help reduce their impact significantly.

Various elements are responsible for the occurrence of these diseases, including age, lifestyle, genetic makeup, blood pressure, glucose level, and cholesterol. However, if these diseases go unnoticed in the early stages, they may cause serious health problems.

The diagnosis of diseases relies on physical examinations, lab tests, and expertise from clinicians. While the traditional method of diagnosing ailments has proved effective, it may take some time before diseases are identified.

In recent times, there has been considerable development in machine learning techniques for analyzing vast amounts of medical data. This approach involves identifying trends within complex relationships between health indicators, thus predicting emerging diseases.

The proposed work involves the design of a system known as HealthRiskAI which is used for analyzing medical datasets and making predictions on various diseases. Datasets of diseases like diabetes, heart disease, and kidney disease are used to develop machine learning models that make appropriate predictions.

The aim of this study includes:

- Analyzing healthcare datasets through data science methods
- Recognizing correlations between medical features
- Developing models for predicting various diseases
- Designing a website for health risk prediction

This study shows the use of machine learning methods in the field of healthcare to help diagnose various diseases.

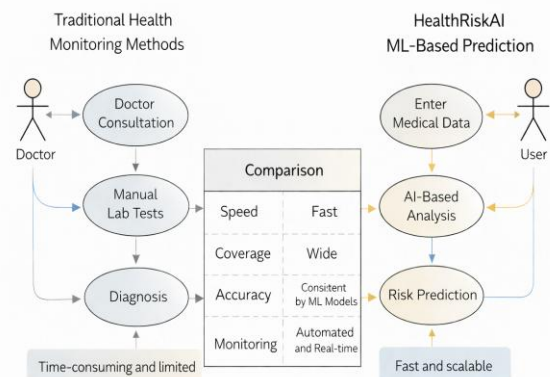


Fig. 1: Flowchart illustrating the comparison between conventional health diagnosis methods and the proposed HealthRiskAI machine learning–based prediction system.

2. LITERATURE REVIEW

Several authors have researched health risk prediction and disease diagnosis using diverse medical and computational approaches. Initially, most studies utilized clinical approaches, where physicians diagnosed diseases such as diabetes, cardiovascular complications, and renal disorders using medical histories, physical examination results, and laboratory findings.

With technological advancements, the use of electronic health records (EHR) and data-intensive techniques has become prevalent to predict health risks and facilitate disease diagnosis. Researchers employ EHR and data-intensive approaches to capture, process, and analyze large amounts of medical data for early diagnosis and decision-making purposes.

Some recent studies have analyzed the effectiveness of various machine learning models in predicting diseases. Such models include decision trees, random forests, support vector machines (SVM), and neural networks. Machine learning models identify complex relationships among health attributes such as age, blood pressure, blood sugar, and cholesterol levels for reliable and timely health risk prediction.

Some commonly used public health datasets include the Pima Indian Diabetes dataset, heart disease dataset, and chronic kidney disease dataset. The datasets contain useful medical attributes that researchers use to train and evaluate machine learning models for health care applications.

Fig. 2: UML use case diagram depicting user interactions within the HealthRiskAI system boundary

3. Dataset Description

The three sets of data used in this analysis include the following:

- (i) Chronic Kidney Disease dataset
- (ii) Heart Disease dataset
- (iii) Diabetes dataset

Each set is composed of the various attributes related to each illness or disease. The Chronic Kidney Disease (CKD) dataset is made up of medical data from patients who have CKD. These include the following attributes: age, blood pressure, specific gravity, albumin, sugar, blood glucose random, blood urea, and serum creatinine.

Each of these attributes serves as an important indicator of CKD, which is used to detect CKD in its early stages.

The Heart Disease dataset comprises some of the attributes associated with the cardiovascular system, including age, sex, type of chest pain, resting blood pressure, cholesterol, maximum heart rate achieved, fasting blood sugar, and exercise-induced angina.

The Diabetes dataset (Diabetes dataset (Pima Indians Diabetes)) includes variables like Glucose, Body Mass Index (BMI), Insulin, Skin Thickness, Number of Pregnancies, Diabetes Pedigree Function, and Age. All these variables are considered crucial for the diabetes prediction process.

Among the list of available variables in both datasets, a selection of relevant variables is made considering their significance medically, availability in data, and contribution to performance measures. Table I presents the selected variables along with their processing methods.

The raw data did not have any common risk categorization for all diseases. Hence, target variables for different diseases were defined separately as follows:

- Kidney Disease: category (ckd / notckd)
- Heart Disease: target (disease present/ absent)
- Diabetes: Outcome (diabetic / non-diabetic)

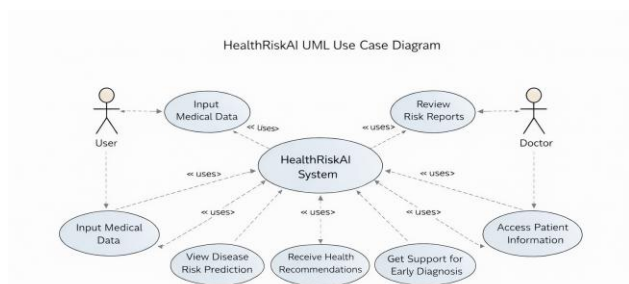


Table I: Selected Features and Preprocessing

Feature Type	Preprocessing
Categorical Features (e.g., sex, cp)	Label Encoding
Numerical Features (e.g., age, glucose, cholesterol)	Normalization
Missing Values (Categorical)	Imputed with 'unknown'
Missing Values (Numerical)	Mean/Median Imputation
Missing Values (Numerical)	Mean/Median Imputation

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All the target variables were encoded to binary classes (0 or 1).

Some attributes had missing values. The missing categorical values were substituted with 'unknown'. Missing numeric values were dealt with using mean or median substitution methods to avoid any loss of data.

The datasets were then preprocessed for better compatibility with machine learning models. The categorical variables were numerically encoded using Label Encoding. Normalization was applied on numeric features to enhance their predictive power.

4. Methodology

The development process of HealthRiskAI involves an organized machine learning approach consisting of:

4.1 Data Acquisition

The datasets used in the current investigation were sourced from publicly available databases of healthcare institutions in CSV format. Three different datasets were used, including Chronic Kidney Disease dataset, Heart Disease dataset, and Diabetes dataset. These are actual datasets of patients' health records and are commonly used by researchers working with predictive modeling tasks.

4.2 Data Preparation

For the preparation of datasets, a Python program was written using libraries Pandas and NumPy to perform various operations on them.

The preprocessing process consisted of:

- Missing value handling via mean/median imputation
- Removal of unnecessary attributes from datasets
- Conversion of categorical attributes to numerical values using Label Encoding method
- Scaling of features using normalization techniques

4.3 Exploratory Data Analysis (EDA)

Exploratory data analysis was done on the datasets to determine the underlying pattern of distribution of the datasets. Visual techniques were employed to find the patterns and relationships within the health parameters.

Visualizations included:

- Heatmaps for correlations
- Histograms for analyzing feature distributions
- Count plots for categorical variables

Through these visualizations, key features were identified and the impacts that various medical attributes have on disease prediction were analyzed.





Fig. 3: Visualizations according to the input data of the patients.

4.4 Machine Learning Model

The machine learning model was used to train the datasets and make predictions about the occurrence of the diseases in the patient samples.

A random forest algorithm was chosen to perform the classification for the prediction of all three diseases. This is a machine learning technique that uses a combination of different decision trees to improve the accuracy of predictions.

The diseases predicted included:

- Prediction of kidney disease
- Prediction of heart disease
- Prediction of diabetes

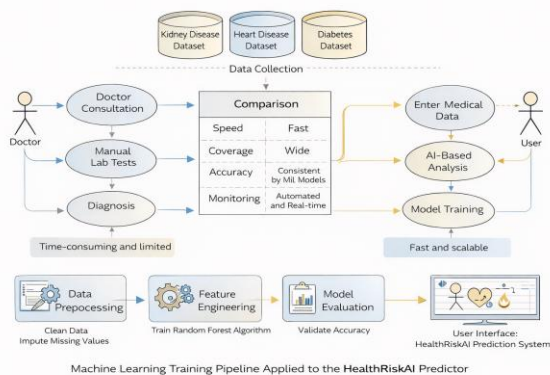


Fig. 3: Machine learning training pipeline applied to the Health Risk AI predictor

4.5 Prediction Pipeline

A prediction pipeline was designed to support real-time predictions of diseases from user input. Once the user inputs his/her medical information, the following process happens:

- (1) The user enters his/her health features, including age, glucose levels, blood pressure, etc., using the input interface.
- (2) The entered data is forwarded to the backend server.
- (3) The data is preprocessed, which includes scaling and encoding.
- (4) The corresponding model that has been trained (kidney, heart, or diabetes) is chosen.
- (5) The model predicts whether there is risk associated with the disease (0/1).
- (6) The prediction result is then presented to the user.

This pipeline facilitates timely and efficient prediction of health risk.

4.6 Development of Web Application

A web application named "HealthRiskAI" was developed to provide an interaction platform where users can submit their input and receive predictions.

The components used in building the web interface are as follows:

- HTML (for creating webpage structure)
- CSS (for adding style)
- Python (using Flask framework)

4.7 Model Performance Evaluation

The performance of the developed models was tested based on their accuracy levels. The outcomes are presented in Table 2 below.

Model	Disease	Accuracy
Random Forest	Kidney Disease	95%
Random Forest	Heart Disease	88%
Random Forest	Diabetes	82%

Table 2: Machine Learning Models' Accuracy Levels

Based on the outcomes of the evaluation, it is evident that the models have attained reasonable accuracy levels in predicting the associated diseases.

5. System Flow

User Input → Data Preprocessing → Feature Selection → Model Prediction → Output Result

6. Results and Analysis

The models designed for the prediction of kidney disease, heart disease, and diabetes performed effectively in terms of identifying possible health risks.

In the case of kidney disease prediction, the classifier accurately sorted patients depending on their medical indicators like blood pressure, glucose, and serum creatinine levels. The model worked efficiently and accurately in terms of prediction.

In the case of heart disease prediction, medical factors including cholesterol level, heart rate, and chest pain classification became crucial elements for prediction. The model performed effectively in predicting heart disease risk in patients.

In the case of diabetes prediction, the model predicted diabetes risk accurately based on the patient’s glucose level, body mass index, and insulin.

The system integrates multiple machine learning models into a single interface, making it efficient and scalable for real-world healthcare applications.



Figure 3: Main Landing Page of HealthRiskAI System

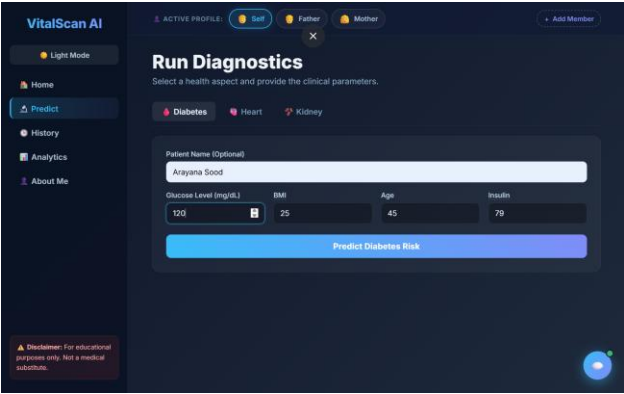


Figure 4: Diabetes Disease Input Interface



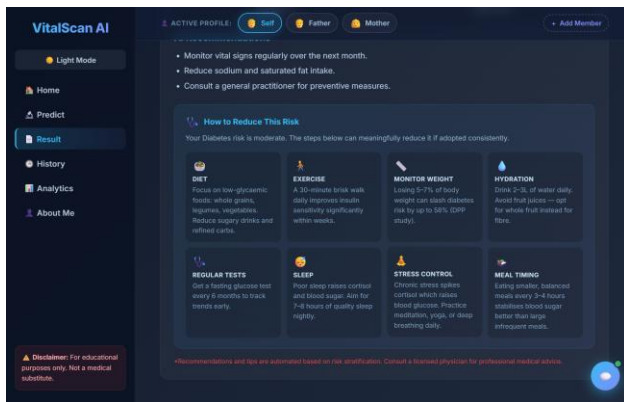


Figure 5: Diabetes Disease Prediction Output

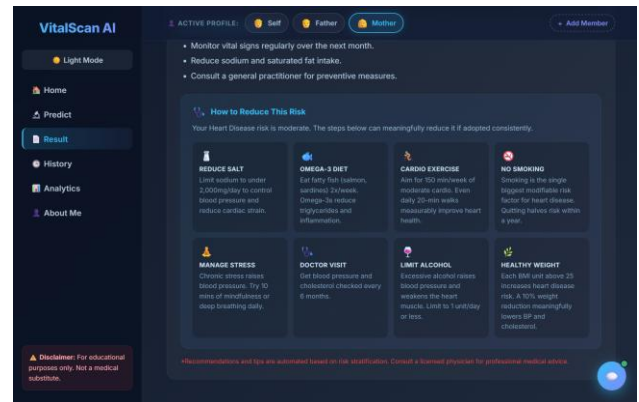


Figure 7: Heart Disease Prediction Output

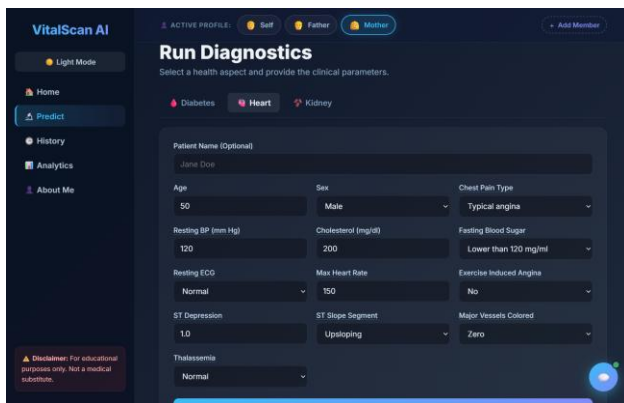


Figure 6: Heart Disease Input Interface

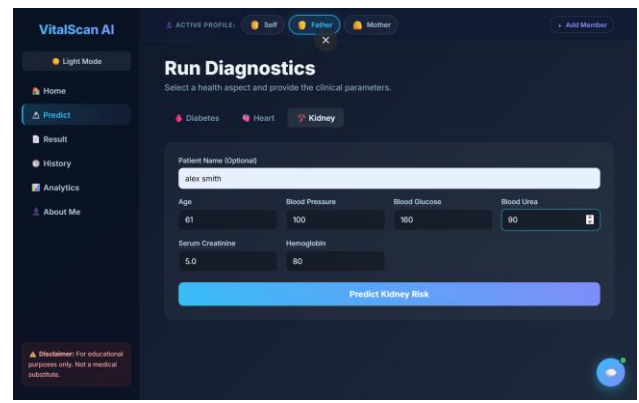
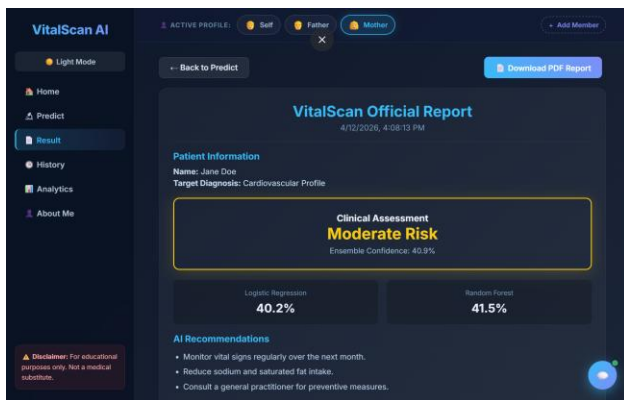


Figure 8: Kidney Disease Input Interface



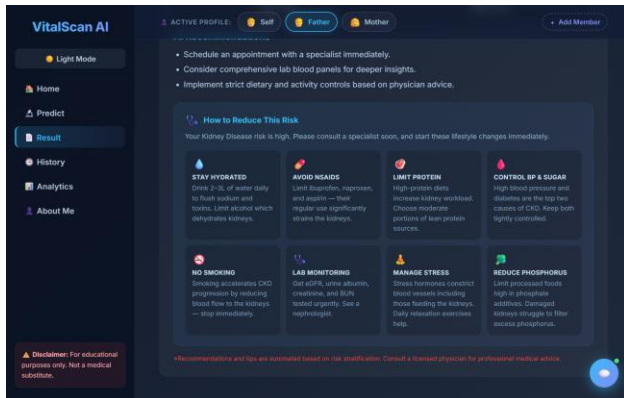


Figure 9: Kidney Disease Prediction Output

6. Future Scope

The proposed system can be improved by using larger and more diverse datasets to enhance model accuracy and performance. Additional diseases such as liver disorders and cancer risk can also be included to make the system more comprehensive.

Advanced techniques like deep learning and ensemble methods can be explored for better prediction results. Integration with real-time data from wearable devices and hospital systems can further improve the reliability of predictions.

The system can also be deployed as a web or mobile application to increase accessibility. In the future, personalized health recommendations can be added to help users take preventive measures based on the predicted risks.

- Integration with real-time hospital data
- Deployment as a web/mobile application
- Inclusion of more diseases
- Use of deep learning models

7. Conclusion

This paper proposes the development of HealthRiskAI, which is a multidisease prediction system created using machine learning approaches to predict whether a person will develop kidney disease, heart disease, and diabetes. For this purpose, such techniques as data preprocessing, features selection, and classification have been used to predict a patient's health condition based on his input data.

The analysis has proven the effectiveness of the trained prediction models and the reliability of their work. The usage of the Random Forest algorithm helped achieve higher prediction accuracy. The combination of several disease prediction models into one makes it more efficient and user-friendly.

The system offers the possibility to enter medical data and get predictions in return, which helps make it applicable in practice. Thus, the importance of using artificial intelligence technologies for solving health problems becomes more evident.

However, there are ways to improve the proposed model. First, it should be based on bigger data sets, and it is necessary to integrate real-time medical data. Moreover, a new version of this system might be developed as a web or mobile app. To conclude, it is worth noting that this approach shows its applicability and efficiency in practical terms.

References

1. D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, School of Information and Computer Sciences, 2017. [Online]. Available: <https://archive.ics.uci.edu/ml/index.php>
2. Python Software Foundation, "Python Documentation," 2023. [Online]. Available: <https://docs.python.org/3/>
3. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
4. Project Jupyter, "Jupyter Notebook Documentation," 2023. [Online]. Available: <https://jupyter.org/documentation>
5. L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.